LLM surprisal as a meaningful indicator of writing proficiency in L2 French

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LTRC 2026 · Montréal

Can we have both?

What is linguistic surprisal anyway ?

Surprisal

How unexpected was the word?

"The cats chased the **mice**" → low

"The cats chased the **thunder**" → high

$$-\log_2 P(word \mid context)$$ (Hale, 2001; Levy, 2008)

Higher values = text less aligned with target-language expectations

Surprisal “spikes” when writing diverges from what the model expects

Sample sentence from the Resorts corpus. Background shading = surprisal score (white: 0 bit; red: 15.9 bits, 90th percentile).
LLM: CamemBERTv2 (masked token model)

* ‘This report considers the views suffered by the participants in the survey distributed last year’

- consider the (prejudice? damage?) suffered by the participants
- consider the views (expressed? shared?) by the participants

What does surprisal correlate with?

Domain	GB English	FR French
Readability	✓ Blache & Rauzy, 2011; Klein et al., 2025	✓ Blache & Rauzy, 2011; Hernandez et al., 2022
Cognitive processing during reading	✓ Fernandez Monsalve et al., 2012; Michaelov et al., 2024; Wilcox et al., 2023;	✓ Ivchenko & Grabar, 2025; Ivchenko et al., 2025
Spoken production & prosody	✓ Clark et al., 2025; Malisz et al., 2018	✓ Liang et al., 2023
Clinical markers	✓ Flick & Ostrand, 2025; Rezaii et al., 2023	✓ Fraser et al., 2019
L2 writing proficiency	✓ Naismith et al., 2022; Cong, 2024; Cong, 2025; Sánchez et al., 2024; Yang et al, 2012	❓ **The current study**
Linguistic anomalies (e.g. spelling / grammar errors)	✓ Li et al., 2021; Sánchez et al., 2024	

Good to know: Cong (2025) and Hu & Cong (2025) also report results for Chinese writers.

Two research questions

RQ1

Agreement with human ratings

Does document-level surprisal **differ across human-rated proficiency levels?**

RQ2

Covariation with error incidence

Does surprisal **covary with independent indicators** of the construct, i.e., incidence of errors in specific categories?

Data & analyses

Two L2 French corpora

- *Test du Certificat de Compétence en Langue Seconde*, U. Ottawa (Appel et al. 2023)
- **Resorts:** 234 argumentative essays (M = 425 words)
- **Smoking:** 166 descriptive essays (M = 391 words)
- Human ratings: **Lower** (≈ A2+ in CEFR) vs **Higher** (≈ B2–C1). Balanced distribution.

Surprisal scoring

- ALSI/ILSA toolkit (Loignon, 2021)
- Large language model (LLM): CamemBERTv2 (Antoun et al., 2024)
- Sentence context

Error identification

- Antidote 12
- Categories: spelling, grammar, syntax, “other”

Lower-proficiency essays show *higher* surprisal

Document-level means in bits; CamemBERTv2. All differences p < .001.

Corpus	Surprisal		r	Cohen's d
	Lower M (SD)	Higher M (SD)		
Resorts	3.56 (0.57)	2.82 (0.44)	−.59	−1.46
Smoking	2.82 (0.53)	2.17 (0.39)	−.57	−1.39

Single feature classifies essays with 75–78% accuracy

Logistic regression with single predictor; bootstrapped (1,000 reps) 95% CI.

Corpus	Cohen's κ	Accuracy	ROC AUC
Resorts	.51 [.37, .67]	.76 [.68, .84]	.85 [.78, .91]
Smoking	.55 [.38, .72]	.78 [.69, .86]	.85 [.77, .92]

Appel et al. (2023), Resorts corpus, 8 NLP features: **Accuracy = .72**

Error incidence increases with surprisal

*Incidence rate ratios per 1-SD increase (negative binomial GLMM). *** p < .001*

Error category	Incidence rate ratio	
	Resorts	Smoking
Spelling	1.71 *** *[1.57, 1.86]*	1.70 *** *[1.55, 1.87]*
Grammar	1.42 *** *[1.31, 1.53]*	1.50 *** *[1.37, 1.64]*
Syntax	1.45 *** *[1.30, 1.61]*	1.55 *** *[1.36, 1.77]*
Other	1.16 *** *[1.06, 1.27]*	1.29 *** *[1.16, 1.44]*

spelling > grammar = syntax > other

Discussion

What we found

- **Agreement:** good correlation between surprisal and human ratings
- **Error covariation:** higher surprisal tracks different categories of errors

Implications for Automated Essay Scoring (AES)

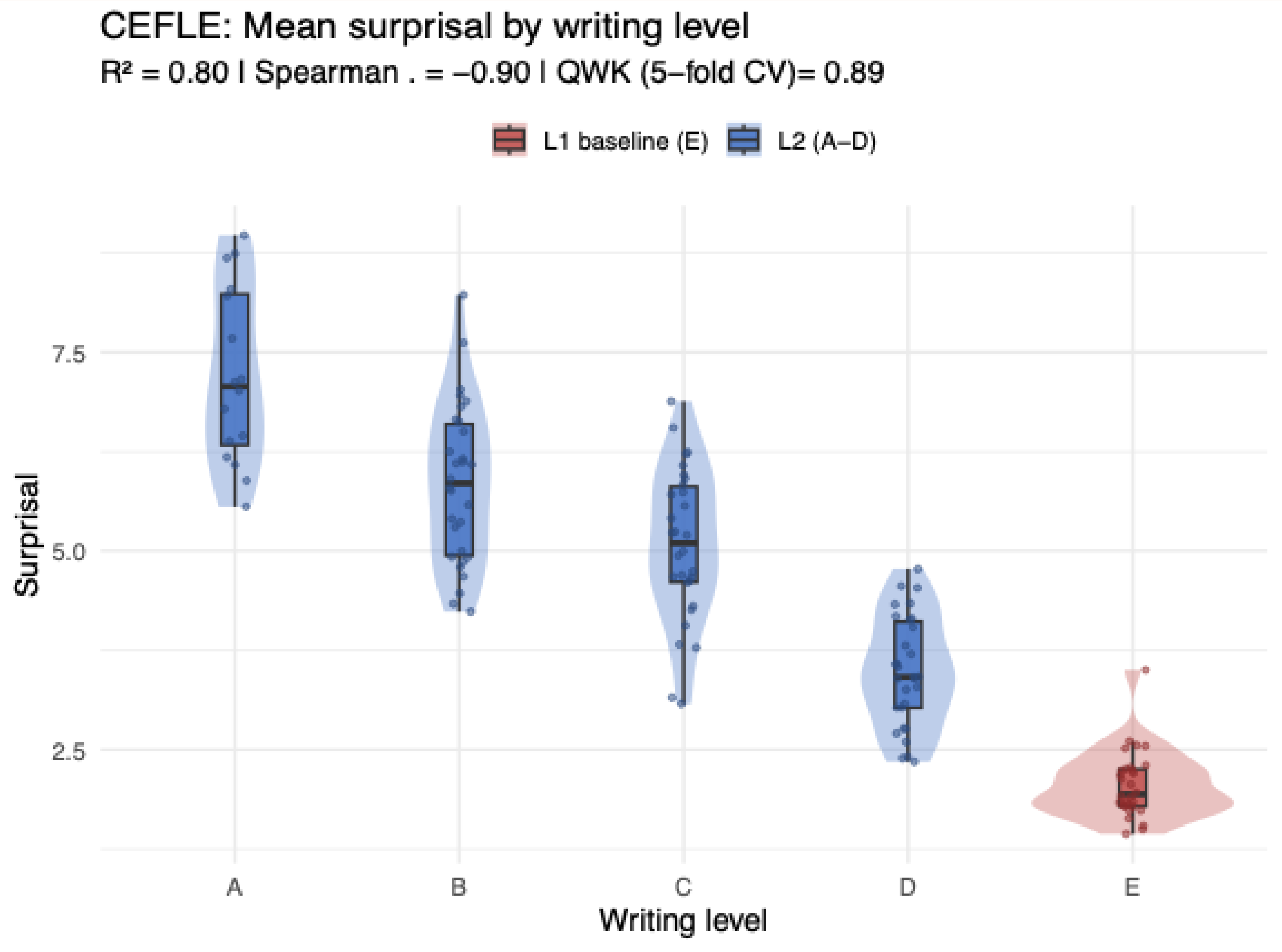
- Extends current evidence to L2 French
- Bridges interpretability and predictive power

Limitations / Next steps

- Test on a corpus with more proficiency levels
- Control for learner characteristics
- Word-level analyses would improve error analyses
- How does French makes surprisal “spikes” compared to other languages: open question

Try it at home! Replication on the CEFLE corpus

(Granfeldt et al., 2006) (n = 136)



Surprisal: R^2 = .80

Direkt Profil: R^2 = .735 (Granfeldt & Ågren, 2014)

Scan to reproduce the CEFLE analysis

github.com/gloignon/alsi/

→ demos/demo_cefle_surprisal.R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By Kevin Tremblay
May 23, 2024 • 11:47 AM • 23

THE TRICK:
1. Get essay
2. Get LLM surprisal
3. Profit

LLM SURPRISAL
$= -\log_2 P(\text{next word})$
Higher = worse writing (usually)

MONTREAL — A local father and education professor claims he can now distinguish strong L2 writing from weak L2 writing using a single number.
Critics say the approach is "oversimplified."
Supporters note that it might help resurrect NLP after all.

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Cong (2025) and Hu & Cong (2025) also report results for Chinese L1 writers.

Merci! Thank you

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Run your own LLM surprisal / entropy analyses

References

Antoun, W., et al. (2024). CamemBERT 2.0: A smarter French language model aged to perfection. *arXiv:2411.08868.*

Appel, R., Arias, A., Baker, B., & Loignon, G. (2023). Insights from lexical and syntactic analyses of a French for academic purposes assessment. *Assessing Writing, 58*, 100789.

Blache, P., & Rauzy, S. (2011). Predicting linguistic difficulty by means of a morpho-syntactic probabilistic model. In *Proceedings of PACLIC 25* (pp. 160–167). Waseda University.

Clark, T. H., Burchill, Z., Florian, T., Jaeger, T. F., & Levy, R. (2025). The relationship between surprisal and prosodic prominence in conversation reflects intelligibility-oriented pressures. *Cognitive Science, 49*, e70134.

Cong, Y. (2024). Second language learning of degree expressions: A computational approach. *Natural Language Processing, 31*, 1187–1209.

Cong, Y. (2025). Demystifying large language models in second language development research. *Computer Speech & Language, 89*, 101700.

Fernandez Monsalve, I., Frank, S. L., & Vigliocco, G. (2012). Lexical surprisal as a general predictor of reading time. In *Proceedings of EACL 2012* (pp. 398–408). ACL.

Ferrara, S., & Qunbar, S. (2022). Validity arguments for AI-based automated scores. *Journal of Educational Measurement, 59*(3), 288–313.

Fraser, K. C., Linz, N., Li, B., Fors, K. L., Rudzicz, F., König, A., ... & Kokkinakis, D. (2019, June). Multilingual prediction of Alzheimer's disease through domain adaptation and concept-based language modelling. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers) (pp. 3659-3670).

Flick, G., & Ostrand, R. (2025). Automatically calculated context-sensitive features of connected speech improve prediction of impairment in Alzheimer's disease. *Journal of Speech, Language, and Hearing Research, 68*(11), 5341–5362.

Granfeldt, J., Nugues, P., et al. (2006). CEFLE and Direkt Profil: A new computer learner corpus in French L2 and a system for grammatical profiling. *LREC-2006 Proceedings*, 565–570.

Granfeldt, J., & Ågren, M. (2014). SLA developmental stages and teachers' assessment of written French: Exploring Direkt Profil as a diagnostic assessment tool. *Language Testing, 31*(3), 285–305.

Hale, J. (2001). A probabilistic Earley parser as a psycholinguistic model. *NAACL 2001 Proceedings.*

Hernandez, N., Oulbaz, N., & Faine, T. (2022). Open corpora and toolkit for assessing text readability in French. In *Proceedings of READI @ LREC 2022* (pp. 54–61). ELRA.

Hu, J., & Cong, Y. (2025). Modeling Chinese L2 writing development: The LLM-surprisal perspective. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics* (pp. 172–183). ACL.

Ivchenko, O., & Grabar, N. (2025). A French eye-tracking corpus of original and simplified medical, clinical, and general texts (FETA). In *Proceedings of Gaze4NLP @ ACL 2025* (pp. 26–51).

Ivchenko, O., Qazi, A. M., & Nasir, J. A. (2025). Where patients slow down: Surprisal, uncertainty, and simplification in French clinical reading. In *Proceedings of Gaze4NLP @ ACL 2025* (pp. 52–57).

Klein, K. G., Frenkel, S., Shubi, O., & Berzak, Y. (2025). Readability Formulas, Systems and LLMs are Poor Predictors of Reading Ease. arXiv preprint arXiv:2502.11150.

Levy, R. (2008). Expectation-based syntactic comprehension. *Cognition, 106*(3), 1126–1177.

Li, B., Zhu, Z., Thomas, G., Xu, Y., & Rudzicz, F. (2021). How is BERT surprised? Layerwise detection of linguistic anomalies. In *Proceedings of ACL-IJCNLP 2021* (pp. 4215–4228).

Liang, Y., Amsili, P., & Burnett, H. (2023). Uniformité de la densité informationnelle: le cas du redoublement du sujet. In *Actes de CORIA-TALN 2023* (pp. 164–176). ATALA.

Loignon, G. (2021). ILSA: an automated language complexity analysis tool for French. *Mesure et évaluation en éducation, 44*(special), 61–88.

Malisz, Z., Brandt, E., Möbius, B., Oh, Y.-M., & Andreeva, B. (2018). Dimensions of segmental variability: Interaction of prosody and surprisal in six languages. *Frontiers in Communication, 3*, 25.

Naismith, B., Mulcaire, P., & Burstein, J. (2022). The University of Pittsburgh English Language Institute Corpus (PELIC). *International Journal of Learner Corpus Research, 8*(1), 55–81.

Rezaii, N., Michaelov, J., Josephy-Hernandez, S., Ren, B., Hochberg, D., Quimby, M., & Dickerson, B. C. (2023). Measuring sentence information via surprisal: Theoretical and clinical implications in nonfluent aphasia. *Annals of Neurology, 94*, 647–657.

Sánchez, M. R., Dobnik, S., & Volodina, E. (2024). Harnessing GPT to study second language learner essays: Can we use perplexity to determine linguistic competence? In *Proceedings of BEA 2024*, 414–427.

Wilcox, E. G., Pimentel, T., Meister, C., Cotterell, R., & Levy, R. P. (2023). Testing the predictions of surprisal theory in 11 languages. *Transactions of the Association for Computational Linguistics, 11*, 1451–1470.

Yang, M.-C., Kim, M.-J., Lee, H.-G., & Rim, H.-C. (2012). Assessing writing fluency of non-English-speaking students for automated essay scoring. *CSEDU 2012 Proceedings.*